

CStats - Lab #3

March 10, 2026

Exercise 1.

Vitamin B₆ is one of the vitamins in a multiple vitamin pill manufactured by a pharmaceutical company. The pills are produced with a mean of 50 mg of vitamin B₆ per pill. The company believes that there is a deterioration of 1 mg/month, so that after 3 months it expects that $\mu = 47$. A consumer group suspects that $\mu < 47$ after 3 months.

(a) Define null hypothesis H_0 and alternative hypothesis H_1 and then a critical region to test H_0 against H_1 at an $\alpha = 0.05$ significance level based on a random sample of size $n = 20$.

(b) If the 20 pills yielded a mean of $\bar{x} = 46.94$ with a standard deviation of $s = 0.15$, what is your conclusion?

(c) What is the approximate p -value of this test?

Solution 1.

(a)

- $H_0 : \mu = 47$
- $H_1 : \mu < 47$

Since the population standard deviation is unknown, we use the t -statistic which is written by $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$. It follows a t -distribution with $n - 1 = 19$ degrees of freedom (df) under H_0 . To determine whether the hypothesis test should be left-tailed, right-tailed, or two-tailed, we analyze the alternative hypothesis H_1 . We have the following instructions:

- **Right-Tailed:** $\mu > k$ (It means μ is more than the value k stated in H_0 .)
- **Left-Tailed:** $\mu < k$ (It means μ is less than the value k .)
- **Two-Tailed:** $\mu \neq k$ (It means μ is different from the value k .)

Thus, we use left-tailed test for this problem. The t -distribution is symmetric about zero, which means we are interested in $\mathbb{P}(T < -c) = \mathbb{P}(T > c) = 0.05$ with $df = 19$. Thus, the critical value c for a left-tailed test at $\alpha = 0.05$ and $df = 19$ is $t_{\alpha,19} = t_{0.05,19} = -1.729$ from t -table. Thus, the critical region is $t < -1.729$.

(b) Let compute the test statistic by t -test. Then we can write the following:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{46.94 - 47}{0.15/\sqrt{20}} = \frac{-0.06}{0.03354} \approx -1.79$$

We already know the critical region is $t < -1.729$ from part (a). Then, since $-1.79 < -1.729$, we reject H_0 at $\alpha = 0.05$, concluding that there is significant evidence to suggest that $\mu < 47$.

(c) t -table provides critical values for positive t -values and their corresponding probabilities as we mentioned before ($\mathbb{P}(T < -c)$). Then, we need to estimate the value $p = \mathbb{P}(T < -1.79)$. From t -table, we have $\mathbb{P}(T < -1.729) = 0.05$ and $\mathbb{P}(T < -1.833) = 0.04$. Therefore, we can say that $P(T < -1.79) \approx 0.044$. Since $p = 0.044 < 0.05$, we reject H_0 , which confirms our results in the previous parts.

Exercise 2.

A company that manufactures motors receives reels of 10,000 terminals per reel. Before using a reel of terminals, 20 terminals are randomly selected to be tested. The test is the amount of pressure needed to pull the terminal apart from its mate. This amount of pressure should continue to increase from test to test as the terminal is "roughed up." (Since this kind of testing is destructive testing, a terminal that is tested cannot be used in a motor.) Let W equal the difference of the pressures: "test No. 1 pressure" minus "test No. 2 pressure." Assume that the distribution of W is $N(\mu_W, \sigma_W^2)$.

(a) Define the null hypothesis and alternative hypothesis for the given problem.

(b) Give the test statistic and a critical region that has a significance level of $\alpha = 0.05$. Sketch a figure illustrating this critical region.

(c) Use the data in Table 1 to calculate the value of the test statistic, and state your conclusion clearly.

(d) What would the conclusion be if $\alpha = 0.01$?

(e) What is the approximate p -value of this test?

Terminal	Test 1	Test 2	Terminal	Test 1	Test 2
1	2.5	3.8	11	7.3	8.2
2	4.0	3.9	12	7.2	6.6
3	5.2	4.7	13	5.9	6.8
4	4.9	6.0	14	7.5	6.6
5	5.2	5.7	15	7.1	7.5
6	6.0	5.7	16	7.2	7.5
7	5.2	5.0	17	6.1	7.3
8	6.6	6.2	18	6.3	7.1
9	6.7	7.3	19	6.5	7.2
10	6.6	6.5	20	6.5	6.7

Table 1: Terminal-Test Data

Solution 2.

(a) We want to test whether the pressure required to pull the terminal apart increases from the first test to the second test. Since the terminal is expected to become "roughed up," we hypothesize that the pressure required should increase, meaning W should be negative on average.

- The null hypothesis $H_0 : \mu_W = 0$ (There is no the difference between pressures for the first and second test.)
- The alternative hypothesis $H_1 : \mu_W < 0$ (The first test requires less pressure than the second, supporting the expectation that the pressure required increases with roughing up.)

(b) We define W as the difference in pressures such that $W_i = (\text{Test 1})_i - (\text{Test 2})_i$. Given that $W \sim N(\mu_W, \sigma_W^2)$, we conduct a one-sample t -test with null hypothesis $H_0 : \mu_W = 0$ and alternative hypothesis $H_1 : \mu_W < 0$. The test statistic (t -test) is $t = \frac{\bar{W}-0}{s_W/\sqrt{n}}$ where $\bar{W}$ is the sample mean of the differences, s_W is the sample standard deviation and $n = 20$ is the number of paired observations. From the t -table for 19 degrees of freedom and $\alpha = 0.05$, $t_{0.05,19} = -1.729$. Thus, the critical region is $t < -1.729$, which means that if our computed test statistic is below -1.729 , we reject H_0 .

(c) Let compute the differences as the following:

$W_1 = 2.5 - 3.8 = -1.3,$	$W_{11} = 7.3 - 8.2 = -0.9,$
$W_2 = 4.0 - 3.9 = 0.1,$	$W_{12} = 7.2 - 6.6 = 0.6,$
$W_3 = 5.2 - 4.7 = 0.5,$	$W_{13} = 5.9 - 6.8 = -0.9,$
$W_4 = 4.9 - 6.0 = -1.1,$	$W_{14} = 7.5 - 6.6 = 0.9,$
$W_5 = 5.2 - 5.7 = -0.5,$	$W_{15} = 7.1 - 7.5 = -0.4,$
$W_6 = 6.0 - 5.7 = 0.3,$	$W_{16} = 7.2 - 7.5 = -0.3,$
$W_7 = 5.2 - 5.0 = 0.2,$	$W_{17} = 6.1 - 7.3 = -1.2,$
$W_8 = 6.6 - 6.2 = 0.4,$	$W_{18} = 6.3 - 7.1 = -0.8,$
$W_9 = 6.7 - 7.3 = -0.6,$	$W_{19} = 6.5 - 7.2 = -0.7,$
$W_{10} = 6.6 - 6.5 = 0.1,$	$W_{20} = 6.5 - 6.7 = -0.2.$

The sample mean difference is $\bar{W} = \frac{-6.8}{20} = -0.34$ and the sample standard deviation is $s_W = \sqrt{\frac{1}{n-1} \sum (W_i - \bar{W})^2} \approx 0.607$. Then, test statistic by t -test is written by

$$t = \frac{-0.34}{0.607/\sqrt{20}} = \frac{-0.34}{0.1358} \approx -2.50.$$

Since $t = -2.50$ is in the critical region ($t < -1.729$), we reject H_0 . At $\alpha = 0.05$, there is sufficient evidence to conclude that $\mu_W < 0$, meaning that the pressure required in Test 1 is lower than in Test 2.

(d) For $\alpha = 0.01$ and 19 degrees of freedom, $t_{0.01,19} = -2.539$. Since $t = -2.50$ is not less than -2.539 , we fail to reject H_0 , which means we accept H_0 . At $\alpha = 0.01$, there is not enough evidence to conclude that $\mu_W < 0$.

(e) From the t -table, $P(T < -2.539) \approx 0.01$, $P(T < -2.093) \approx 0.025$. Since $t = -2.50$ is between these values, the approximate p -value is $p \approx 0.015$ because it should be closer to 0.01. Then, we conclude that since $p = 0.015 < 0.05$, we reject H_0 at $\alpha = 0.05$, but since $p = 0.015 > 0.01$, we fail to reject H_0 at $\alpha = 0.01$.

Exercise 3.

To measure air pollution in a home, let X and Y equal the amount of suspended particulate matter (in $\mu\text{g}/\text{m}^3$) measured during a 24-hour period in a home in which there is no smoker and a home in which there is a smoker, respectively. We shall test the null hypothesis $H_0 : \sigma_X^2/\sigma_Y^2 = 1$ against the one-sided alternative hypothesis $H_1 : \sigma_X^2/\sigma_Y^2 > 1$.

(a) If a random sample of size $n = 9$ yielded $x = 93$ and $s_x = 12.9$ while a random sample of size $m = 11$ yielded $y = 132$ and $s_y = 7.1$, define a critical region and give your conclusion if $\alpha = 0.05$.

(b) Now test $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X < \mu_Y$ if $\alpha = 0.05$.

Solution 3.

(a) We use F-test to decide the critical region.

$$F = \frac{s_x^2}{s_y^2} = \frac{(12.9)^2}{(7.1)^2} = \frac{166.41}{50.1} \approx 3.32. \quad (1)$$

Here are the degrees of freedoms: $df_x = 8$ and $df_y = 10$. From the F-table, we have $F_{0.05}(8, 10) \approx 3.07$. Thus, critical region is $F > 3.07$. We conclude that we reject H_0 .

(b) Because variances are not equal in part (a), we should use the Welch two-sample t-test.

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}} = \frac{-39}{\sqrt{23.07}} \approx -8.13. \quad (2)$$

Moreover, since variances are not the same, we use the following formula to calculate degrees of freedom:

$$df = \frac{\left(\frac{s_x^2}{n} + \frac{s_y^2}{m}\right)^2}{\frac{(s_x^2/n)^2}{n-1} + \frac{(s_y^2/m)^2}{m-1}} \approx 12 \quad (3)$$

For a left-tailed test with $\alpha = 0.05$, we have $t_{0.05}(12) = -1.782$. The critical region is $t < -1.782$. So, since $-8.13 < -1.782$, we reject H_0 .

Exercise 4.

Consider the computer output shown in Figure 1.

Two-Sample T-Test and CI: Y1, Y2				
Two-sample T for Y1 vs Y2				
	N	Mean	Std. Dev.	SE Mean
Y1	20	50.19	1.71	0.38
Y2	20	52.52	2.48	0.55
Difference = mu (X1) - mu (X2)				
Estimate for difference: -2.33341				
95% CI for difference: (-3.69547, -0.97135)				
T-Test of difference = 0 (vs not =): T-Value = -3.47				
P-Value = 0.001 DF = 38				
Both use Pooled Std. Dev. = 2.1277				

Figure 1: Two-Sample t-test

- Can the null hypothesis be rejected at the 0.05 level? Why?
- Is this a one-sided or a two-sided test?
- If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$ would you reject the null hypothesis at the 0.05 level?
- If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 < 2$ would you reject the null hypothesis at the 0.05 level? Can you answer this question without doing any additional calculations? Why?
- Use the output and the t table to find a 95 percent upper confidence bound on the difference in means.
- What is the p -value if the hypotheses are $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$?

Solution 4.

(a) Yes, because $p = 0.001 < 0.05 = \alpha$.

(b) This is a two-sided test since it tests the equality of two means where $H_0 : \mu_1 = \mu_2$ against $H_A : \mu_1 \neq \mu_2$.

(c) Assuming same variances (since the degrees of freedom $df = n_1 + n_2 - 2 = 38$), T-test is calculated by

$$t = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad (4)$$

where $S_p^2 = \frac{(n_1-1)S_{Y_1}^2 + (n_2-1)S_{Y_2}^2}{n_1+n_2-2}$. Thus, we have $S_p = \sqrt{4.53725} \approx 2.13$, which is already given in the output. Then, when we calculate it for $\mu_1 - \mu_2 = 2$,

$$t \approx -6.44 \quad (5)$$

From t-table, we have $t_{0.05/2,38} \approx -2.024$. Since $-6.44 < -2.024$, we reject H_0 .

(d) Yes, it is enough to look at t-table. This is one-tailed test (left-tailed), so we are interested in the critical region $t < t_{0.05,38} \approx -1.69$. Since $-6.44 < -1.69$ we reject H_0 .

(e) It is already given in the output, but here is the formula:

$$(\bar{Y}_1 - \bar{Y}_2) \mp t_{\alpha,df} \cdot SE \quad (6)$$

where SE is pooled standard deviation.

(f) This is two-tailed test. From part c, we have $t = -6.44$. Then, using t-table, we look for $t = -6.44$ with $df = 38$; thus, we conclude with $p < 0.0005$.